Engine Model(s): PT6A

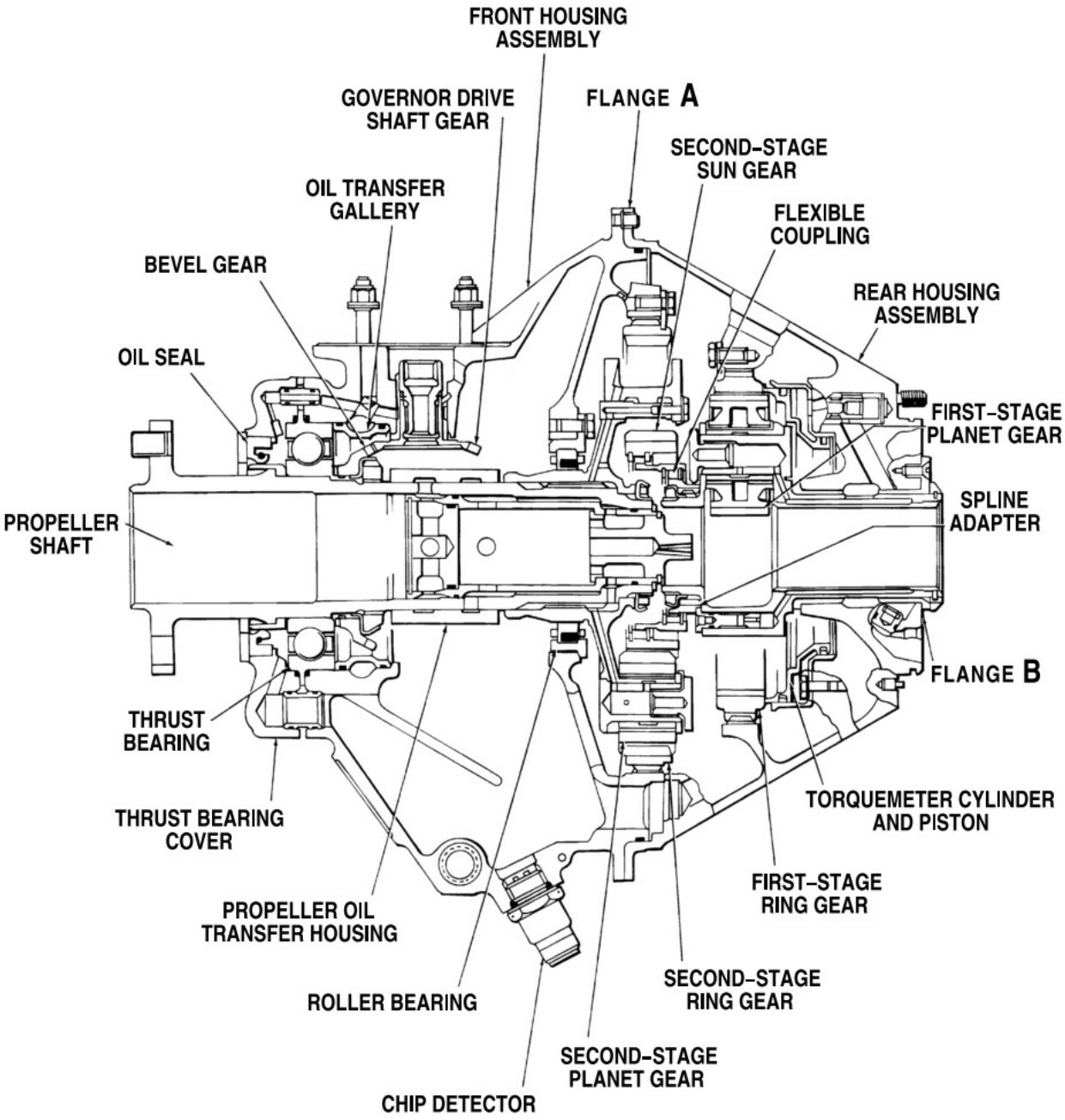
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REDUCTION GEARBOX - DESCRIPTION AND OPERATION

1. Description and Operation (Ref. Fig. 1)

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Figure 1 Reduction Gearbox Assembly - Cross-section



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A. General

The engine models included in this manual are equipped with one of two propeller reduction gear and shaft sections, that differ primarily in pitch of first-stage reduction gears.

Both sections have a nominal output shaft speed of 2200 rpm at takeoff power; one is fitted with coarse first-stage gear teeth and the other with fine first-stage gear teeth for increased torque.

The reduction gear and propeller shaft, located at the front of the engine, is housed in two magnesium alloy castings, the front and rear cases of which are bolted together at flange A of the exhaust duct (Ref. 72-50-05).

The gearbox contains a two-stage planetary geartrain, accessory drives, torquemeter and propeller shaft. The first-stage reduction gear and torquemeter are contained in the rear case, while the second-stage reduction gear, accessory drives and propeller shaft are in the front case. Torque from the power turbine section is transmitted through the power turbine shaft and turbine coupling to the first-stage sungear.

The first-stage sungear consists of a short hollow steel shaft which has an integral spur gear at the front end and an external spline on the rear end. The external splines engage with the internal splines of the turbine coupling on the power turbine shaft and is secured within the coupling by two retaining rings.

The first-stage ring gear is located in helical splines provided in the rear case of the reduction gearbox. The torque developed by the power turbine is transmitted through the sungear and planet gears to the ring gear which is opposed by the helical splines. This opposition causes the planet gear carrier to rotate. The ring gear, although secured by the helical splines, is allowed to move axially between the case and five retaining plates secured to the case. The axial movement is used in the torquemeter application.

The second-stage of reduction is housed in the front case of the reduction gearbox. The first-stage planet carrier is attached to the second-stage sungear by a flexible coupling which also acts to dampen any vibrations between the two rotating masses.

The second-stage sungear drives five planet gears mounted in the second-stage carrier. A second-stage ring gear is located by splines in the front case of the reduction gearbox by retaining plates. The second-stage carrier is in turn splined to the propeller shaft and held by a keywasher and nut. A flanged roller bearing supports the second-stage carrier and the rear of the propeller shaft. The inner race of the bearing is located on the machined circumference of the second-stage carrier extension, while the flanged outer race is bolted to a web in the front case.

A propeller oil transfer tube adapter and a nozzle assembly are coupled together by a retaining ring and located within the centerbore of the propeller shaft. The adapter provides the necessary separation of pressure oil for propeller actuation, and pressure oil for lubrication of propeller thrust and gearbox front roller bearings. The nozzle assembly and transfer tube conduct pressure oil for lubrication of the No. 4 bearing area, second-stage planetary gears and sungear coupling. Pressure oil is supplied to the oil transfer tube via oil transfer sleeve installed on the OD of the propeller shaft. The sleeve is supplied by two independent oil sources via two

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transfer tubes. Engine oil is supplied to the rear section of the sleeve for direction to the centerbore transfer tube adapter. Propeller governor oil is supplied to the front section of the sleeve to provide hydraulic pressure for the propeller pitch servo system (Ref. 76-10-00).

The accessories located on the front reduction gearbox are driven by a bevel drive gear mounted at the rear of the propeller shaft thrust bearing assembly. Drive shafts from the bevel drive gear transmit rotational power to the three pads on the front case of the reduction gearbox. The pads are located at the 12 o'clock, and approximately the 3 and 9 o'clock positions.

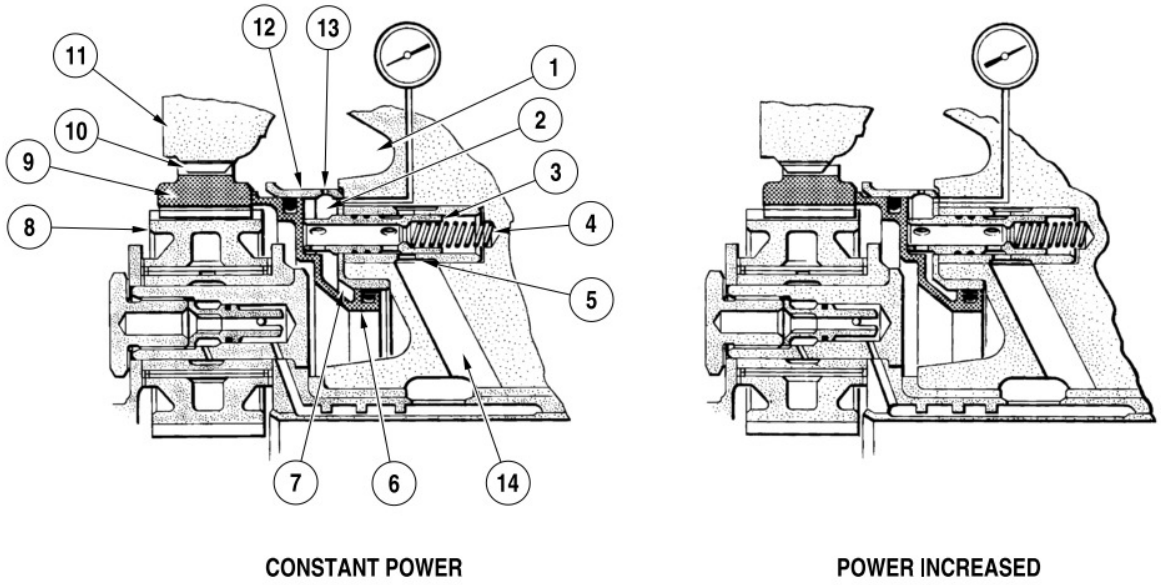
Propeller thrust loads are absorbed by the ball bearing assembly located on the front section of the propeller shaft, the outer flange of which is secured to the front case of the gearbox. The bevel drive gear adjusting spacer, thrust and roller bearings and seal runner are stacked and secured to the propeller shaft by a keywasher and spanner nut.

A thrust bearing cover assembly is secured by bolts at the front flange of the reduction gearbox front case. The cover assembly incorporates an oil seal retaining ring plate which facilitates replacement of the oil seal and two oil transfer tubes. The tubes allow pressure oil in the reduction gearbox to be applied to the front face of the thrust bearing and scavenge oil to return to the gearbox via cored passages in the cover assembly.

The torquemeter (Ref. Fig. 2) is a hydro-mechanical torque-sensing device, installed at the rear of the first-stage reduction gear in the rear case of the reduction gearbox. The torquemeter provides an accurate indication of engine power output and consists of a cylinder, piston, seal rings, valve plunger and spring.

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Figure 2 Torquemeter – Schematic (Typical)



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1. Gearbox Oil Pressure
2. Torquemeter Pressure
3. Valve Plunger
4. Spring
5. Metering Orifice
6. Torquemeter Piston
7. Torquemeter Chamber
8. First-stage Planet Gear
9. First-stage Ring Gear
10. Helical Splines
11. Rear Case
12. Torquemeter Cylinder
13. Bleed Orifice
14. Engine Oil Pressure

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Rotation of the ring gear is resisted by the helical splines, which impart an axial movement to the ring gear and to the torquemeter piston. Movement of the piston moves a valve plunger against the spring, opening a metering orifice and allowing an increased flow of pressure oil into the torquemeter chamber. Piston movement continues until oil pressure in the torquemeter chamber is proportional to torque being absorbed by the ring gear. Any change in engine power recycles the sequence until a state of equilibrium is again reached.

Hydraulic locks are prevented by allowing oil to bleed continuously from the torquemeter chamber into the reduction gear sump through a small bleed orifice at the top of the torquemeter cylinder.

Because the external ambient pressure and the pressure within the reduction gear may vary and affect the total pressure on the torquemeter piston, internal pressure is measured. The difference between the torquemeter pressure and reduction gear internal pressure accurately indicates the torque being produced. The two pressures are internally routed to bosses at the top of the reduction gear front case where connections can be made to suit individual airframe instrumentation requirements.

A chip detector is installed at the 6 o'clock position on the reduction gearbox. The chip detector provides an indication of the presence of ferrous particles in the lubrication system when a continuity check is done. Connection provisions exist for airframe wiring to provide in-flight indication of contamination (Ref. Aircraft Maintenance Manual).Inspection paint stripes on detector housing permit visual indication of defects of swaging at the electrical connector.

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